Parul® University
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NAAC GRADE A++

Information and Communication Technology

UNIT - 2
DATA PREPROCESSING

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Contents

- Introduction to Data Preprocessing
- Reading
- Selecting
- Filtering Data - Filtering Missing Values
- Manipulating
- Sorting
- Grouping
- Rearranging
- Ranking Data

Introduction to Data Preprocessing

Data Preprocessing is one of the most important steps in the Data Science life cycle. It involves preparing **raw data for analysis by cleaning, transforming** and **organizing it into a suitable format**. Since real-world data is often **incomplete, inconsistent, noisy** or **inaccurate**, preprocessing helps improve data quality before it is used for analysis or machine learning and the quality of data directly affects the accuracy of results.

Introduction to Data Preprocessing

Data Preprocessing is the **process of cleaning, transforming, integrating** and **organizing raw data** to make it suitable for **analysis, visualization** and **predictive modeling.**

Why is Data Preprocessing Important?

Raw data often contains:

1. Improves data quality and reliability.
2. Removes errors, duplicates, and inconsistencies.
3. Handles missing values and outliers.
4. Enhances the performance of machine learning algorithms.
5. Reduces processing time and improves decision-making

If these issues are not handled properly, they can lead to inaccurate analysis and poor model performance.

1. Improves data quality and reliability

Raw data may contain **incorrect**, **irrelevant** or **inconsistent** information. Data pre-processing cleans and organizes the data, making it more accurate and reliable for analysis.

Example: Correcting spelling mistakes in customer names or standardizing date formats.

2. Removes Errors, Duplicates and Inconsistencies

Datasets often contain **duplicate records** or **incorrect values** that can lead to **misleading results**. Pre-processing identifies and removes these issues.

Example: If a customer appears twice in a sales database, one duplicate entry can be removed to avoid incorrect sales calculations.

3. Handles Missing Values and Outliers

Missing data and **unusual values** (**outliers**) can negatively affect analysis and machine learning models. Pre-processing techniques help manage these problems.

Example:

- Missing age values can be replaced with the average age.
- An outlier such as a salary of ₹10 crore in a dataset of average employees can be investigated and corrected if necessary.

4. Enhances the Performance of Machine Learning Algorithms

Machine learning models work best with **clean** and **properly formatted data**. Pre-processing helps algorithms learn patterns more effectively, leading to better predictions.

Example: Scaling features such as age (20–60) and income (₹20,000–₹2,00,000) ensures that one feature does not dominate the model.

5. Reduces Processing Time and Improves Decision-Making-

Clean and **organized data** requires less computational effort, making analysis faster and more efficient. **Better-quality data** also leads to more accurate business decisions.

Example: A hospital can quickly analyse patient records to identify disease trends and improve treatment planning.

Flow of Data Preprocessing-

Raw Data
↓
Data Cleaning
↓
Data Integration
↓
Data Transformation
↓
Data Reduction
↓
Normalization / Standardization
↓
Data Splitting
↓
Ready for Analysis

Steps in Data Pre-processing-

1. Data Collection

Gather data from various sources such as **databases**, **websites**, **sensors**, **surveys** and **social media platforms**.

Steps in Data Pre-processing-

2. Data Cleaning

Data cleaning **removes inaccurate**, **incomplete** or **irrelevant data**.

Tasks include:

- Handling **missing** values
- Removing **duplicate** records
- Correcting **inconsistencies** or **wrong data**
- Detecting and treating **outliers**

Example:

If a student's age is entered as 250 years, it is an error and should be corrected or removed.

Steps in Data Pre-processing-

2. Data Cleaning

Techniques used:

- ✓ **Mean Imputation:** Replaces missing values with the average of the attribute.
- ✓ **Median Imputation:** Replaces missing values with the middle value, useful when outliers exist.
- ✓ **Mode Imputation:** Replaces missing values with the most frequent value.
- ✓ **Deletion Method:** Removes records that contain missing values.
- ✓ **Duplicate Detection:** Identifies and removes repeated records.

Steps in Data Pre-processing-

3. Data Integration

Combining data from multiple sources into a unified dataset. It can be challenging due to differences in data **formats**, **structures** and **schema**.

- Used when data comes from databases, files
- Removes redundancy between datasets
- Resolves conflicts in data values

Example:

Merging customer information from sales, marketing and support databases.

Steps in Data Pre-processing-

3. Data Integration

Techniques used:

- **Schema Matching:** Aligns attributes from different data sources.
- **Entity Resolution:** Identifies records that refer to the same real-world entity.
- **Correlation Analysis:** Finds and removes redundant attributes.
- **Data Conflict Resolution:** Resolves inconsistencies in units or data values.
- **Duplicate Elimination:** Removes overlapping records after integration.

Steps in Data Pre-processing-

4. Data Transformation

Data transformation **converts data into a suitable form** so that data mining algorithms can work effectively.

- Bring data into a common format
- Improve mining efficiency
- Make data suitable for modeling

Steps in Data Pre-processing-

4. Data Transformation

Data transformation **converts data into a suitable form** so that data mining algorithms can work effectively.

Techniques used:

1. Normalization
2. Standardization
3. Aggregation
4. Encoding categorical variables

Example:

Converting "Male" and "Female" into numerical values (0 and 1).

Steps in Data Pre-processing-

Technique	Definition	Simple Example (Before)	After Transformation
Normalization	Rescales numerical values to a common range (usually 0 to 1).	Student Marks: 50, 75, 100	0.0, 0.5, 1.0
Standardization	Converts numerical values so that they have a mean of **0** and a standard deviation of **1**.	Student Heights (cm): 150, 160, 170	-1.22, 0.00, +1.22 (approximately)
Aggregation	Combines multiple records into a summarized value.	Daily Sales: ₹1000, ₹1200, ₹800	Total Sales = ₹3000 or Average Sales = ₹1000
Encoding Categorical Variables	Converts categorical (text) data into numerical values.	Gender: Male, Female, Male	Male = 1, Female = 0 → 1, 0, 1

Steps in Data Pre-processing-

5. Data Reduction-

Reducing the size of data while maintaining its integrity.

Methods include:

I. Feature selection

I. Dimensionality reduction

II. Sampling

Benefits:

- Faster processing
- Reduced storage requirements
- Improved model performance

Steps in Data Pre-processing-

5. Data Reduction-

I. Feature selection

Feature Selection is the process of selecting only the most important features (attributes) from a dataset and removing irrelevant ones.

Real-Life Example: Student Performance Prediction

Original Dataset

Student Name	Roll No	Age	Attendance	Internal Marks	Final Result
A Rahul	101	20	90%	85	Pass

Steps in Data Pre-processing-

5. Data Reduction-

I. Feature selection

For predicting the Final Result, attributes such as Attendance and Internal Marks are important, while Name and Roll Number are usually irrelevant.

After Feature Selection

Age	Attendance	Internal Marks	Final Result
20	90%	85	Pass

Steps in Data Pre-processing-

5. Data Reduction-

II. Dimensionality Reduction

Dimensionality Reduction reduces the number of features while preserving most of the important information.

Real-Life Example: Face Recognition System

A facial image may contain 10,000 pixels (features).

Before

Pixel 1

Pixel 2 Pixel 3

Pixel 10,000

Steps in Data Pre-processing-

5. Data Reduction-

II. Dimensionality Reduction

Using techniques like Principal Component Analysis (PCA), these 10,000 features can be reduced to 100 important features.

After

Feature 1

Feature 2

Feature 100

Steps in Data Pre-processing-

5. Data Reduction-

III. Sampling

Sampling is the process of selecting a representative subset of data from a large dataset.

Real-Life Example: Election Survey

Suppose a country has 100 million voters.

It is difficult to ask every voter for their opinion.

Instead, a survey company selects 10,000 voters from different regions.

Steps in Data Pre-processing-

5. Data Reduction-

III. Sampling

Population: 100,000,000 voters

Sample: 10,000 voters

The survey results are used to estimate the opinion of the entire population.

Benefit: Saves time and cost, Faster analysis, Easier data handling

Applications: Election polls, Customer satisfaction surveys, Market research

Steps in Data Pre-processing-

6. Data Discretization-

Converting continuous data into discrete categories.

Example:

Age:

- 0–18 → Child
- 19–35 → Young Adult
- 36–60 → Adult
- Above 60 → Senior Citizen

Common Data Pre-processing Techniques

Technique	Purpose
1. Missing Value Handling	Fill or remove missing data
2. Data Normalization	Scale data to a common range
3. Standardization	Transform data with mean = 0 and standard deviation = 1
4. Encoding	Convert categorical data into numerical form
5. Outlier Detection	Identify abnormal values
6. Feature Selection	Choose important variables

Data Pre-processing Example

Real-Life Example-

A hospital collects patient data including **age**, **weight**, **blood pressure** and **diagnosis**.

Before predicting diseases:

1. Missing patient records are completed or removed.
2. Duplicate entries are deleted.
3. Data from different hospital departments are merged.
4. Numerical values are standardized.
5. Irrelevant attributes are removed.

Reading

Reading data is the process of importing data from different sources into a software tool or programming environment for analysis.

Data can be stored in various formats such as: CSV files, Excel files, Databases, JSON files, Text file

Reading

Purpose

- Load data into the system
- View and understand dataset structure
- Check data quality
- Begin exploratory analysis

Reading

Example

Suppose a student dataset is stored in an Excel file.

Roll No	Name	Marks
101	Rahul	85
102	Priya	90
103	Aman	78

Reading the file allows us to access and analyze this information.

Reading

```
import pandas as pd


data = pd.read_csv("students.csv")
print(data)
```

Selecting Data

Selecting data means **choosing specific rows, columns** or **portions** of **a dataset** that are relevant to the analysis.

Instead of using the entire dataset, analysts often **focus only on the required information**.

Purpose

- Reduce unnecessary data
- Focus on important attributes
- Simplify analysis
- Improve efficiency

Selecting Data

Selecting Columns

Example - Dataset:

Name	Marks	City
Rahul	85	Indore
Priya	90	Bhopal

Select only **Name** and **Marks**:

Name	Marks
Rahul	85
Priya	90

```
data[['Name', 'Marks']]
```

Selecting Data

Selecting Rows

Select the first two records:

Name	Marks
Rahul	85
Priya	90

```
data.head(2)
```

Filtering Data

Filtering data is the **process of retrieving only those records** that **satisfy specific conditions**.

It helps **focus on relevant data** and **remove unnecessary** records.

Purpose

- Identify specific groups
- Analyze targeted records
- Improve decision-making
- Discover patterns and trends

Filtering Data

Example 1:

Filter Students with **Marks**

Greater Than 80

Name	Marks
Priya	90
Kamal	55
Aman	41
Rahul	85

Filtered Result:

Name	Marks
Rahul	85
Priya	90

```
data[data['Marks'] > 80]
```

Filtering Data

Example 2:

Filter Students from **Indore**

Name	Marks	City
Rahul	85	Indore
Kamal	55	Pune
Aman	41	Indore
Priya	90	Bhopal

Filtered Result:

Name	Marks	City
Rahul	85	Indore
Aman	41	Indore

```
data[data['City'] == 'Indore']
```

Filtering Data

Filtering with Multiple Conditions

Example: Find students from **Indore** who scored **more than 80 marks**.

Name	Marks	City
Rahul	85	Indore
Kamal	55	Pune
Aman	41	Indore
Priya	90	Bhopal

Filtered Result:

Name	Marks	City
Rahul	85	Indore

```
data[(data['City'] == 'Indore') & (data['Marks'] > 80)]
```

Filtering Missing Values

Missing values are data entries that are **empty**, **unavailable** or **recorded** as **NULL**, **NaN (Not a Number)** or **blank**.

Filtering missing values means **identifying** and **handling** these **incomplete records**.

Why Missing Values Occur

1. Data entry errors
2. Incomplete survey responses
3. System failures
4. Data not collected
5. Database migration issues

Filtering Missing Values

Methods to Handle Missing Values

A. Remove Missing Values

B. Replace Missing Values

Filtering Missing Values

Methods to Handle Missing Values

A. Remove Missing Values

Delete rows or columns containing missing values.

Example

Name	Age	Marks
Rahul	20	85
Priya	21	NULL
Aman	22	90

Filtering Missing Values

Methods to Handle Missing Values

A. Remove Missing Values

After removing missing values:

Name	Age	Marks
Rahul	20	85
Aman	22	90

Filtering Missing Values

Methods to Handle Missing Values

B. Replace Missing Values

Missing values can be replaced using:

I. Mean

II. Median

III. Mode

IV. Constant value

Filtering Missing Values

Methods to Handle Missing Values

B. Replace Missing Values

Example:

If student marks are: 80, 85, **NULL**, 90

Replace **NULL** with average:

Average = (80 + 85 + 90) / 3 = **85**

Final Data: 80, 85, **85**, 90

Filtering Missing Values

Methods to Handle Missing Values

B. Replace Missing Values

```
import pandas as pd

data = pd.read_csv("students.csv")

data = data.dropna()          # Remove missing values
# or
data = data.fillna(data.mean(numeric_only=True))    # Replace with mean
```

Data Manipulation

Data manipulation is the **process of modifying data** to make it suitable for analysis.

It includes:

- ✓ Adding new columns
- ✓ Deleting columns
- ✓ Updating values
- ✓ Renaming columns
- ✓ Creating calculated fields

Data Manipulation

Example: Original Table

Name	Marks
Rahul	85
Priya	90

Add Grade column:

Name	Marks	Grade
Rahul	85	A
Priya	90	A+

```
data["Grade"] = ["A", "A+"]
```

Sorting Data

Sorting means **arranging data in a specific order**.

Types of sorting:

1. Ascending Order
2. Descending Order

Sorting Data

Example: Original Data

Name	Marks
Rahul	85
Aman	70
Priya	95

Ascending Order

Name	Marks
Aman	70
Rahul	85
Priya	95

Descending Order

Name	Marks
Priya	95
Rahul	85
Aman	70

```
data.sort_values(by="Marks", ascending=False)
```

Grouping Data

Grouping is the **process of combining records** that **have similar values** so that **aggregate calculations** can be performed.

Common calculations:

1. Sum
2. Average
3. Count
4. Maximum
5. Minimum

Grouping Data

Example:

Department	Salary
HR	40000
IT	60000
HR	45000
IT	65000

Group by Department:

Department	Average Salary
HR	42500
IT	62500

```
data.groupby("Department")["Salary"].mean()
```

Rearranging Data

Rearranging means **changing the order of rows** or **columns** to improve **readability** and **analysis**.

Example: Original Order

NAME	AGE	MARKS

Rearranged Order

NAME	MARKS	AGE

This makes analysis easier depending on the requirement.

```
data = data[["Name", "Marks", "Age"]]
```

Ranking Data

Ranking **assigns a position** or **rank** to **each record** based on a **particular** value.

Ranking is useful for identifying:

- ✓ Highest scorer
- ✓ Best-performing employee
- ✓ Top-selling products

Ranking Data

Example:

Name	Marks
Rahul	95
Aman	88
Priya	72

Rank Table

Name	Marks	Rank
Rahul	95	1
Aman	88	2
Priya	72	3

```
data["Rank"] = data["Marks"].rank(ascending=False)
```

Reference Books

1. **Data Science for Business** – Foster Provost and Tom Fawcett, *Data Science for Business*, O'Reilly Media, 2013.
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